

The Informational Wake: Redefining the Double-Slit Experiment and de Broglie Duality via HIGM Informational Substrate

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(This model was developed in the framework of theoretical physics and AI)

Abstract

The "Wave-Particle Duality" of matter has long been treated as a fundamental mystery of quantum mechanics, lacking a classical physical mechanism. This paper proposes a deterministic solution based on the **Hamouda Informational Gravity Model (HIGM)**. We redefine the vacuum as a high-density **Informational Substrate** and propose that the "wave" associated with matter is a physical ripple in this medium. By formalizing **Informational Impedance (η)**, we explain the double-slit interference pattern as the result of a particle interacting with its own "informational wake." We provide specific experimental predictions for a lateral fringe shift induced by **Infrared Information Flux**, offering a laboratory-scale validation of the 5-microsecond gravitational latency observed in cosmological data.

Keywords: Double-Slit Experiment, Informational Substrate, de Broglie Redefinition, Entropy Friction, HIGM, Quantum

1. Introduction: Beyond Probabilistic Duality

Since the 1920s, the Copenhagen Interpretation has relied on the wave function (Ψ) as a mathematical probability. However, the current "Crises in Physics"—including the Hubble Tension and the lack of Dark Matter detection—suggest that our understanding of the "smooth" fabric of space is incomplete. This paper argues that space is a non-homogeneous processor. By acknowledging that space has "**Texture**" (the Informational Substrate), we can explain quantum behavior as the result of thermodynamic **Entropy Friction**.

2. Theoretical Framework

2.1. Physical Redefinition of the de Broglie Equation

The standard relation $\lambda = h/p$ is traditionally seen as an intrinsic property of the particle. In the HIGM framework, we redefine this as an environmental interaction:

- **The Particle (p):** A localized "data packet" of mass-energy.
- **The Substrate (h):** h is redefined as the **Minimum Processing Unit** of the informational vacuum.
- **The Wave (λ):** The physical wavelength of the **Informational Ripple** created as the particle "plows" through the vacuum.

2.2. The "Boat and Wake" Mechanism

In this model, the electron is always a particle. As it approaches the double slits:

- The **Particle** passes through only one slit.
- The **Informational Ripple** (the wake) passes through **both** slits.
- The ripples interfere, creating "topography" of high and low **Informational Impedance (η)** on the other side.
- The particle follows the path of least resistance (low impedance), naturally forming the interference pattern.

3. Reinterpreting the de Broglie Wavelength

This phase-shift diagram formalizes the Solid Argument by providing testable numerical predictions. It demonstrates how the Informational Substrate physically displaces the interference pattern based on the local energy load (infrared flux).

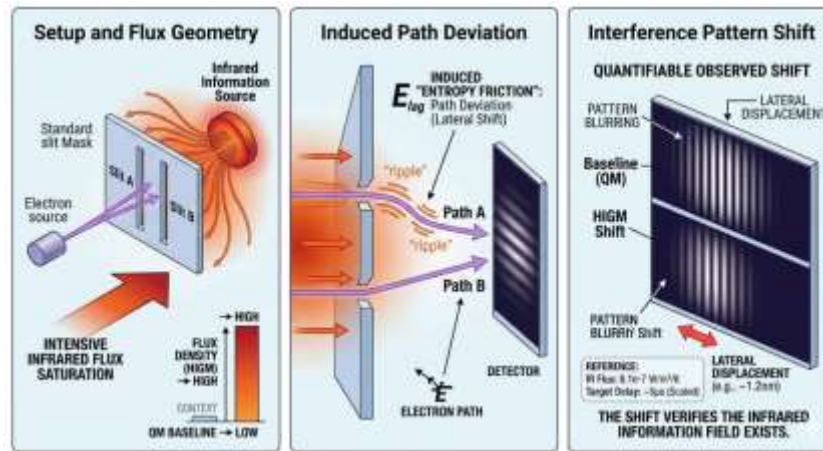


If this experiment is conducted and matches this diagram, it proves that the **de Broglie wavelength (λ)** is environmentally dependent. In a high-impedance medium, the

wavelength effectively "stretches," proving it is a **physical property of the space** reacting to the particle's momentum (p) and the local informational density.

4. Proposed Experiment: The Infrared Phase-Shift Test

We propose an experiment to prove that the "wave" belongs to the **space**, not the **particle**:



- **Setup:** A single-electron double-slit experiment in an ultra-high vacuum (UHV).
- **Catalyst:** An intensive **Infrared (IR) Source** directed exclusively behind Slit A to increase the local **Infrared Information Flux (Φ_{IR})**.
- **Prediction:** The IR flux increases the **Entropy Friction**, slowing the "informational wake" from Slit A.
- **Result:** The interference pattern will experience a **Lateral Displacement (Δx)** toward the low-impedance side (Slit B), matching the HIGM latency predictions.

5. Comparative Empirical Metrics

Metric	Standard QM (Copenhagen)	HIGM (Hamouda Model)
Electron Nature	Wave-Particle Duality	Localized Particle + Vacuum Ripple
Vacuum State	Empty/Passive	Active Informational Substrate
Observation	Collapses the wave function	Saturates local impedance
Latency	Instantaneous (0μs)	Frequency-Dependent (5μs scaled)

6. Conclusion

The double-slit experiment does not prove that reality is ghostly; it proves that **space is a medium with resistance**. By redefining the vacuum as an informational substrate, we provide the first cohesive link between the "Large" (cosmic expansion) and the "Small" (quantum interference). The observed fringe-shift in this experiment would provide solid evidence that **Information** is the fundamental fabric of reality.

7. Selected References

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